

How it works

Markdown → HTML → Puppeteer (Chromium) → PDF

The generator renders a branded HTML page and prints it to PDF using Puppeteer with `displayHeaderFooter: true`. Header and footer templates are rendered by Chromium into the dedicated `@page` margin zones — completely separate from the content area, guaranteeing no overlap on any page.

For multi-page documents, two separate PDF passes are merged with `pdf-lib`: one for page 1 (with logo), one for pages 2+ (without logo).

Key files

File	Purpose
<code>generate-eureka-pdf.mjs</code>	Entry point: renders HTML, runs PDF passes, merges output
<code>generate-eureka-brand-header.mjs</code>	Builds the header template (logo + grey tile)
<code>generate-eureka-brand-footer.mjs</code>	Builds the footer template (contact info + orange blob)
<code>generate-eureka-brand-content-styles.mjs</code>	CSS for layout and markdown content
<code>eureka-document-config.mjs</code>	All sizing constants
<code>eureka-document-pdf-structure.html</code>	Base HTML shell with content slot

Sizing constants (`eureka-document-config.mjs`)

Constant	Purpose
<code>headerHeightPx</code>	Height of the header band — used as Puppeteer <code>margin.top</code>
<code>footerHeightPx</code>	Height of the footer band — used as Puppeteer <code>margin.bottom</code>
<code>defaultPagePaddingPx</code>	Uniform padding applied to content and used for logo/footer alignment
<code>a4PageWidthPx</code>	Full page width in px — used to size templates correctly
<code>chromiumTemplateBodyMarginPx</code>	Chromium injects <code>body { margin: 8px }</code> into templates — this constant (8) is used to compensate

Header/footer template system

Puppeteer's `displayHeaderFooter: true` renders `headerTemplate` and `footerTemplate` into the `@page` margin areas. The content area is fully protected on every page.

1. **Count pass** — render full document with logo header, count pages
2. If single page → write directly (reuse count buffer)
3. If multi-page:
 - **Page 1 pass** — `pageRanges: '1'`, `headerTemplate` with logo
 - **Pages 2+ pass** — `pageRanges: '2-'`, `headerTemplate` without logo
 - **Merge** — combine with `pdf-lib`

Conditional logo

The logo (`eureka-logo-horizontal.svg`) is loaded once at startup as a base64 data URI. `buildEurekaHeader({ showLogo: true/false })` controls whether the `<img>` is included in the template.

Constraints

- Chrome must be available at `/usr/bin/google-chrome`
- All assets (logo, HTML shell) must be in the `assets/` directory